

Fieldrove — autonomous targeted spraying for row crops

The Problem

A farmer spraying herbicide across a field treats every square metre to reach the 4% of it that has weeds. The chemical cost is carried on the whole field, the resistance pressure builds across the whole field, and regulators in most markets are tightening application limits every year.

Targeted spraying solves this in principle and has for a decade. In practice the machines that do it either require a tractor operator to drive slowly enough for the vision system to keep up, or cost more than the chemical they save.

Our Solution

Fieldrove is an autonomous spraying unit that identifies individual weeds at 14 km/h and actuates a nozzle within 40 milliseconds.

The vision stack runs on the unit with no connectivity requirement, which matters because the fields we operate in have none. The classifier distinguishes our twelve target weed species from the crop at 96.4% precision in field conditions, including at dusk and in light rain, which is where most published results stop being reproducible.

The unit navigates the field autonomously from a boundary map, operates unsupervised for nine hours on one charge, and returns to its dock.

The Market

Row crop area in North America and Europe is approximately 180 million hectares. Herbicide spend on that area is around \$19B a year.

At our measured reduction, the chemical saving alone is between \$28 and \$61 per hectare per season depending on crop and weed pressure. Regulatory tightening adds to that: three European markets have set reduction targets that conventional broadcast spraying cannot meet.

Competition

Two large equipment manufacturers offer targeted spraying as an attachment to their own tractors, which requires the farmer to own that tractor. Three startups build autonomous units; two operate below 6 km/h, which puts the field capacity below what a commercial farm needs.

Our differentiation is speed with autonomy, and the reason it took four years is that the two are in tension.

The Team

Dr. Mateo Ferreira spent six years in agricultural robotics research and led the perception team on a publicly funded autonomous weeding project. Sanna Korhonen was a mechanical lead at an agricultural equipment manufacturer, responsible for two sprayer product lines.

Twenty-four people, sixteen in engineering.

Traction

- 9 units built
- 3,100 hectares treated across two seasons
- 89% average reduction in herbicide volume, measured against the farmer's prior season on the same fields
- 96.4% weed detection precision in field conditions
- zero crop damage incidents recorded across the treated area

The machine works. That sentence is the result of four years and we are confident in it.

Business Model

We intend to sell the unit outright at around \$180,000, with a software subscription for classifier updates.

At that price the payback for a 1,200-hectare farm is between two and four seasons on chemical saving alone. We have not yet sold a unit at that price. The 3,100 hectares were treated under trial agreements at cost.

Funding

We are raising \$20M to move from nine hand-built units to a production line, expand the classifier to nine further weed species, and run a third season.